Hugo Marquerie 11/03/2025

Lema de Borel-Cantelli I

Lema 1 (de Borel-Cantelli I). *Sea* $(A_n)_{n\in\mathbb{N}} \subset \Sigma$ *en* $(\Omega, \Sigma, \mathbb{P})$ *tal que* $\sum_n \mathbb{P}(A_n) < \infty$

$$\implies \mathbb{P}\left(\limsup_{n\to\infty} A_n\right) = 0.$$

***Demostración*:** Definimos $N \colon \Omega \longrightarrow \mathbb{R} \cup \{\infty\}$ dada por $N(\omega) = \sum_{n\in\mathbb{N}} \mathbb{1}_{A_n}(\omega)$, entonces

$$\limsup_{n\to\infty} A_n = \{\omega \in \Omega : N(\omega) = \infty\}.$$

Por tanto, $\mathbb{E}[N] \overset{\substack{\text{TCM}\\ \downarrow}}{=} \sum_{n\in\mathbb{N}} \mathbb{E}\left[\mathbb{1}_{A_n}\right] = \sum_{n\in\mathbb{N}} \mathbb{P}(A_n) < \infty.$

$$\implies N \in \mathcal{L}^1(\mathbb{P}) \implies \mathbb{P}\left(\{\omega \in \Omega : N(\omega) = \infty\}\right) = 0.$$

Concluimos por tanto que $\mathbb{P}(\limsup A_n) = 0$. ■

Referenciado en

- `teo-convergencia-martingalas`
- `convergencia-probabilidad-imp-subsucesion-casi-segura`
- `prop-borel-cantelli-iii`
- `ley-fuerte-grandes-numeros`

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